Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Google Professional Machine Learning Engineer Training

Course Code: C740

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 8.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
7.0	Legacy release	Original four-day course outline aligned to eight legacy topics.	Tertiary Infotech Academy
8.0	12 September 2026	Rebuilt four-day plan against six current Google PMLE domains, exact 415-slide deck, 16 Activities and original WA/PP timing.	Dr Alfred Ang

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Course Information

Course title	Google Professional Machine Learning Engineer Training
Course code	C740
Duration	4 days · 30 instructional hours (9:30am – 5:30pm daily)
Delivery	Instructor-led technical course with 16 Activities and mock evidence packs

Learning Outcomes and Criteria

- LO1: Assess and draft Google Cloud solution specifications to meet expected business machine learning requirements.
- LO2: Develop implementation plans and resolve issues for Google Cloud machine learning solutions.
- LO3: Develop and review processes for metrics associated with Google Cloud machine learning solution implementation.

Slide and Activity Alignment

Domain	Approx. weight	Deck slides	Activities	Legacy coverage
D1 Architecting low-code AI solutions	~13%	19–83	02, 03, 04, 05	Topics 1–3: GCP fundamentals, Google ML services, ML workflow
D2 Collaborating to manage data and models	~16%	84–148	01, 06, 11	Topics 1, 3 and 5: big data, data quality, feature engineering
D3 Scaling prototypes into ML models	~21%	149–213	07, 08, 09, 10	Topics 4–6: TensorFlow, feature engineering, enterprise ML
D4 Serving and scaling models	~20%	214–278	12, 13	Topics 6–7: batch/online prediction, production systems
D5 Automating and orchestrating ML pipelines	~18%	279–343	14, 15	Topic 8: MLOps, CI/CD and repeatable workflows
D6 Monitoring AI solutions	~13%	344–408	16	Topics 6–8: testing, monitoring, troubleshooting and MLOps

Detailed Four-Day Schedule

Day 1 — Low-code AI architecture, Google Cloud setup and data foundations

Time	Minutes	Mode	Content / deck alignment	Evidence or resource
09:30–09:50	20	Admin	Slides 1–11: welcome, trainer profile,	Slides + LG

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			learner context, ground rules and course facts	
09:50–11:00	70	Concept	D1 slides 19–83: low-code architecture and service selection	Slides + LG
11:00–13:00	120	Practical	Activities 01–02: safe project/account and architecture decision record	Activity folder + LG
13:00–13:30	30	Lunch	Lunch break	Lunch
13:30–15:10	100	Concept	D1: BigQuery ML, AutoML, AI APIs, Gemini and responsible low-code design	Slides + LG
15:10–17:10	120	Practical	Activities 03–05: BigQuery ML, AutoML plan and managed AI APIs	Activity folder + LG
17:10–17:30	20	Review	Evidence review and Q&A	Slides + LG

Total counted duration: 450 instructional minutes (7.5 hours), 9:30am – 5:30pm.

Day 2 — Feature engineering, TensorFlow and scalable custom training

Time	Minutes	Mode	Content / deck alignment	Evidence or resource
09:30–09:45	15	Review	Day 1 retrieval practice	Slides + LG
09:45–11:10	85	Concept	D2 slides 84–148: data modalities, preprocessing, Feature Store and privacy	Slides + LG
11:10–13:00	110	Practical	Activities 06 and 11: data quality, features and	Activity folder + LG

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			point-in-time parity	
13:00–13:30	30	Lunch	Lunch break	Lunch
13:30–14:55	85	Concept	D3 slides 149–213: TensorFlow, custom training, tuning and scaling	Slides + LG
14:55–17:15	140	Practical	Activities 07–09: Keras, Vertex custom training and hyperparameter tuning	Activity folder + LG
17:15–17:30	15	Review	Evidence review and Q&A	Slides + LG

Total counted duration: 450 instructional minutes (7.5 hours), 9:30am – 5:30pm.

Day 3 — Model serving, deployment strategies and production scaling

Time	Minutes	Mode	Content / deck alignment	Evidence or resource
09:30–09:45	15	Review	Day 2 retrieval practice	Slides + LG
09:45–11:10	85	Concept	D3: GPU/TPU/CPU decisions, distributed training and failure modes	Slides + LG
11:10–13:00	110	Practical	Activity 10: distributed PyTorch worker pool and scaling evidence	Activity folder + LG
13:00–13:30	30	Lunch	Lunch break	Lunch
13:30–14:55	85	Concept	D4 slides 214–278: serving patterns, containers, registry and canary	Slides + LG
14:55–17:15	140	Practical	Activities 12–13: registry/canary and Cloud Run model serving	Activity folder + LG
17:15–17:30	15	Review	Evidence	Slides + LG

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			review and Q&A	

Total counted duration: 450 instructional minutes (7.5 hours), 9:30am – 5:30pm.

Day 4 — MLOps automation, monitoring, responsible AI and wrap-up

Time	Minutes	Mode	Content / deck alignment	Evidence or resource
09:30–09:45	15	Review	Day 3 retrieval practice	Slides + LG
09:45–11:15	90	Concept	D5 slides 279–343: pipeline components, Vertex Pipelines and CI/CD/CT	Slides + LG
11:15–13:00	105	Practical	Activities 14–15: compile pipeline and release gates/model CI	Activity folder + LG
13:00–13:30	30	Lunch	Lunch break	Lunch
13:30–15:00	90	Concept	D6 slides 344–405 and wrap 399–406: monitoring, safety, bias and optimization	Slides + LG
15:00–16:45	105	Practical	Activity 16: drift, bias, SLO and incident evidence pack	Activity folder + LG
16:45–17:30	45	Review	Course wrap-up, exam decision checklist, Q&A and course feedback	Slides + LG

Total counted duration: 450 instructional minutes (7.5 hours), 9:30am – 5:30pm.

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